

Zhaoyang (Charles) Zheng

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EDUCATION

University of British Columbia

Bachelor of Science in Computer Science

Sep. 2020 – Dec 2025

Vancouver, BC

- Enrolled in the **UBC Science co-op program**.
- Relevant courses: **data structures and algorithms**, **machine learning**, **statistics**, and **relational databases**.

EXPERIENCE

Computer Science Instructor (CPSC 110 & AI Education)

Youth Global Education

Jan. 2024 – Dec. 2025

Vancouver, BC / Remote

- Taught **CPSC 110 (Systematic Program Design)** to first-year UBC students, emphasizing structured problem decomposition, data-driven design, and program correctness.
- Designed and delivered **Artificial Intelligence** lessons introducing machine learning concepts and real-world applications to pre-university students.
- Supported learners with no prior programming experience through labs, assignments, and small-group conceptual walkthroughs.

Software Engineer Intern

Learning Tech Rover — University of British Columbia

Jan. 2024 – Sep. 2024

Vancouver, BC

- Developed and maintained faculty web pages and course systems supporting recruitment and digital learning platforms.
- Built and optimized custom UI features using **HTML**, **CSS**, and **JavaScript**, improving usability, accessibility, and platform reliability.
- Created digital training content and tutorials, contributing to a **200%** increase in portal user retention and engagement.

Programming Curriculum Developer

Lebo

May. 2023 – Aug. 2023

Shandong, China

- Developed a comprehensive course on **HTML**, presenting idiomatic concepts in a clear and engaging format.
- Pioneered a low-level course on **C**, covering complex topics on **memory management** and **operating systems**.
- Orchestrated a successful showcase of students' final projects with over **100** attendees.

PROJECTS

Heart Disease Predictor — *R, Python, scikit-learn, pandas*

Sep. 2022 – Jan. 2023

- Trained a **KNN model** to accurately predict over **5** types of heart disease on health telemetry data.
- Developed a robust preprocessing pipeline to ensure a consistent format for the data.
- Tuned hyper-parameters with **cross-validation**, resulting in over **90%** validation accuracy in the final model.

Online Video Game — *PHP, Java, Oracle, SQL*

Jun. 2022 – Aug. 2022

- Developed a powerful web-game GUI with **PHP** to serve as a front-end for the service.
- Engineered a scalable **Java** back-end **API** to interface with the front-end and **Oracle** database.
- Designed a robust **SQL schema** to model the domain of the video-game application.
- Stored relevant game-state information in **Oracle** to back the application, ensuring a seamless user experience.

InsightUBC – Campus Analytics Platform — *TypeScript, Node.js, REST API*

Jan. 2025 – May. 2025

- Implemented a full-stack TypeScript system to ingest and persist campus course and room data.
- Built a domain-specific query engine with filtering, aggregation, and multi-key sorting via RESTful APIs.
- Developed a RESTful web server and interactive frontend for dataset exploration.
- Wrote automated unit and integration tests to ensure correctness and reliability of core system components.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, SQL, C++, R, HTML, CSS, PHP
Technologies: Git, JUnit, pandas, NumPy, scikit-learn, WordPress, Photoshop, Camtasia
Language Proficiency: English, Mandarin